

5.2 Energy: GPE & KE

Question Paper

Course	CIEA Level Physics
Section	5. Work, Energy & Power
Topic	5.2 Energy: GPE & KE
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What is the approximate kinetic energy of an Olympic athlete when running at maximum speed during a 100m race?

- A. 400 J
- B. 4000 J
- C. 40 000 J
- D. 400 000 J

[1 mark]

Question 2

A body travelling with a speed of 10 m s^{-1} has kinetic energy 1500 J.

If the speed of the body is increased to 40 m s^{-1} , what is its new kinetic energy?

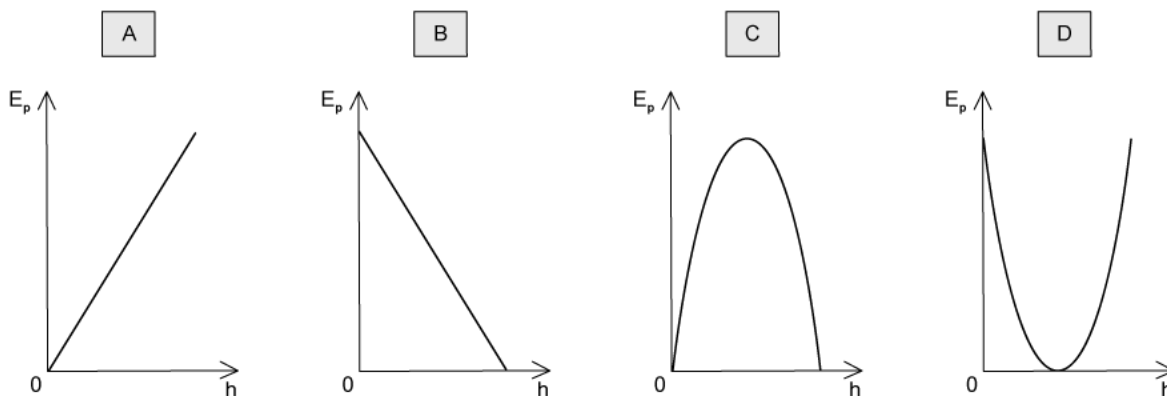
- A. 4.5 kJ
- B. 6 kJ
- C. 24 kJ
- D. 1350 kJ

[1 mark]

Question 3

An object is thrown into the air.

Which graph shows how the potential energy E_p of the object varies with height h above the ground?



[1 mark]

Question 4

A car of mass 1000 kg travels forwards at 25 m s^{-1} and then backwards at 5 m s^{-1}

What is the change in the kinetic energy of the car?

- A. 200 kJ
- B. 300 kJ
- C. 325 kJ
- D. 450 kJ

[1 mark]

Question 5

To get to his office from the entrance of the building, a man has to walk up six flights of stairs.

The height of each flight is 2.5 m and the man has a mass of 80 kg.

What is the approximate gain in the man's gravitational potential energy during the climb?

- A. 1200 J
- B. 2000 J
- C. 4800 J
- D. 12 000 J

[1 mark]

Question 6

A car with a total mass of 1400 kg is travelling at 108 km h^{-1}

What is the kinetic energy of the car?

- A. 21 kJ
- B. 42 kJ
- C. 630 kJ
- D. 8160 kJ

[1 mark]

Question 7

The formula for kinetic energy is $E_k = \frac{1}{2}mv^2$

Which kinematic equation is used in the derivation of this formula?

A. $v = u + at$

B. $s = \left(\frac{u + v}{2}\right)t$

C. $s = ut + \frac{1}{2}at^2$

D. $v^2 = u^2 + 2as$

[1 mark]

Question 8

The formula for gravitational potential energy is $E_p = mgh$

From which concept is this equation derived?

A. Work Done = Force $\times$ Distance

B. Force = Mass $\times$ Acceleration

C. Power = Force $\times$ Speed

D. Distance = Speed $\times$ Time

[1 mark]

Question 9

A cyclist is travelling at a constant speed up a hill. The frictional force resisting the cyclist's motion is 8.0 N .

The cyclist uses 450 J of energy to travel 20 m .

What is the increase in gravitational potential energy of the cyclist?

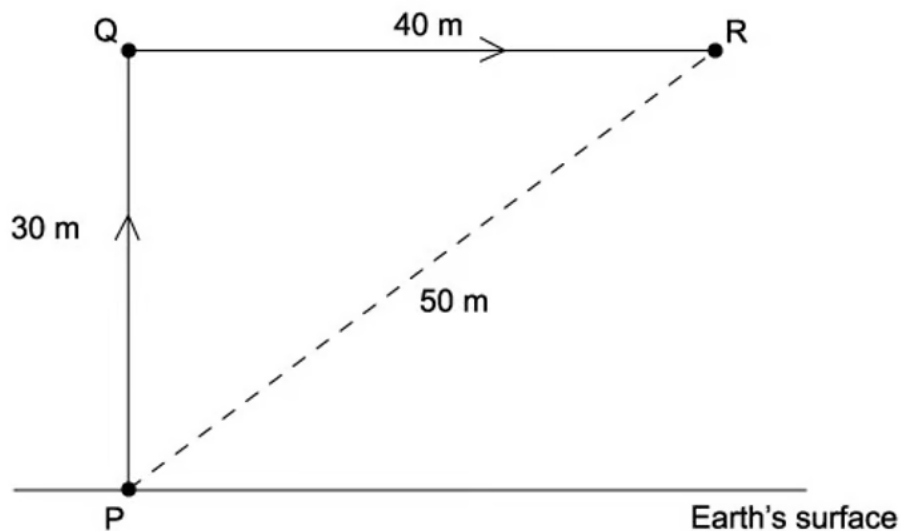
- A. 160 J
- B. 290 J
- C. 440 J
- D. 610 J

[1 mark]

Question 10

A stone of weight 4.0 N in the Earth's gravitational field is moved from P to Q and then to R along the path shown.

How much potential energy does the stone gain?



- A. 120 J
- B. 200 J
- C. 280 J
- D. 1200 J

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- In the first topic we looked at how we can estimate different quantities
- Here we need to estimate the maximum speed and mass of an Olympic athlete
- A reasonable estimate for the time taken to complete a 100 m race in an Olympic competition is about 10 s
- Let the maximum speed of the Olympic athlete be $v = \frac{100}{10} = 10 \text{ m s}^{-1}$
- Let the mass of the Olympic athlete be about 80 kg
 - Kinetic energy $\frac{1}{2}mv^2 = \frac{1}{2} \times 80 \times 10^2 = 4000 \text{ J}$

2

The correct answer is **C** because:

- The kinetic energy of an object is equal to
 - $KE = \frac{1}{2}mv^2$
- The mass of the object is constant, but we must know its value to find the new kinetic energy. Therefore:
- $1500 = \frac{1}{2}m \times 10^2$
- $m = \frac{2 \times 1500}{100} = 30 \text{ kg}$
- Now we have our m and v values, so we can work out the new kinetic energy :
 - $KE = \frac{1}{2} \times 30 \times 40^2 = 24\,000 \text{ J}$
- The new kinetic energy is 24 kJ

3

The correct answer is **A** because:

- The height above the ground is h
- So, as h increases, the position of the object above the ground gets higher
- Potential energy is equal to $E_p = mgh$
- Therefore, as h increases, potential energy E_p increases linearly since the relationship between E_p and h is directly proportional $E_p \propto h$

B is incorrect because this graph shows the object would have more GPE on the ground than when it is a height h above the ground

C & D are incorrect because these graphs show a non-linear increase/decrease in GPE as h increases

4

The correct answer is **B** because:

- The change in kinetic energy is equal to
 - $\Delta KE = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$
- Where $u = 25 \text{ m s}^{-1}$ and $v = 5 \text{ m s}^{-1}$
 - $\Delta KE = \frac{1}{2} \times 1000(25^2 - 5^2) = 300\,000 \text{ J}$
- The change in kinetic energy is 300 kJ

5

The correct answer is **D** because:

- Gravitational potential energy (GPE) = mgh
- The GPE gained by walking up one flight of stairs is $= 80 \times 9.8 \times 2.5 = 1960 \text{ J}$
- The GPE gained by walking up six flights of stairs is $= 6 \times 1960 = 11\,760 \text{ J}$
 - Therefore, the man gains 12 000 J in climbing six flights of stairs

6

The correct answer is **C** because:

- The speed is in km h^{-1} so we must first convert it to m s^{-1}
 - $108 \text{ km h}^{-1} = \frac{108 \times 1000}{60 \times 60} = 30 \text{ m s}^{-1}$
- The kinetic energy is therefore equal to
 - $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 1400 \times 30^2 = 630\,000 \text{ J}$
- The kinetic energy of the car is 630 kJ

7

The correct answer is **D** because:

- In the derivation of kinetic energy (KE), we start by equating KE with the work done by the moving body, then add in Newton's second law of motion $F = ma$
 - $\Delta KE = W = F \Delta s = ma \Delta s$
- Then we want to use a kinematic equation which contains the term $a \Delta s$
 - $v^2 = u^2 + 2a \Delta s$
 - $a \Delta s = \frac{v^2 - u^2}{2}$
 - $\Delta KE = ma \Delta s = m \left(\frac{v^2 - u^2}{2} \right)$
- Now if we expand this equation, we get
 - $\Delta KE = \frac{mv^2}{2} - \frac{mu^2}{2}$
- We assume that the object started at rest and accelerated with a uniform rate of a up to speed v , therefore $u = 0$ and this leaves us with
 - $KE = \frac{1}{2}mv^2$

8

The correct answer is **A** because:

- In the derivation of gravitational potential energy (GPE), we start with the definition of work done
 - $\text{Work done} = \text{force} \times \text{distance} = F \Delta s$
- Next, we consider the case of an object being raised by a height h
- The only force acting on this object will be its weight
 - $F = W = mg$

- To move an object against gravity, the force applied upwards must equal the downwards force gravity exerts on the object, mg . So, if the object is moved against gravity by a distance h the work done is given by :
 - $E_p = mgh$

9

The correct answer is **B** because:

- Since the cyclist is moving at a constant speed, he is not gaining any kinetic energy
- Work done against friction $= F \times d = 8 \times 20 = 160J$
- From the conservation of energy :
- Work done by cyclist = Gain in GPE + Work done against friction
 - $450 = \text{GPE} + 160$
- So, the GPE gained by the cyclist is
 - $\text{GPE} = 450 - 160 = 290 \text{ J}$

10

The correct answer is **A** because:

- Gravitational potential energy (GPE) $= mgh$
- h refers to the vertical distance travelled by the stone, which is 30 m and mg refers to the weight of the stone which is 4 N
 - Therefore, $\text{GPE} = 4 \times 30 = 120 \text{ J}$